

FB₁nference -- generated SCL (B-spline LUT)

```
01 FUNCTION_BLOCK FB_Inference
02 VAR_INPUT  features: ARRAY[0..27] OF REAL;  END_VAR
03 VAR_OUTPUT class_id: INT;  confidence: REAL;  END_VAR
04 FOR i := 0 TO 27 DO
05     lo := 0;
06     FOR j := 1 TO 13 DO
07         IF features[i] >= W_DB.g0[j] THEN lo := j; END_IF;
08     END_FOR;
09     t_val := (features[i]-W_DB.g0[lo]) /
10             (W_DB.g0[lo+1]-W_DB.g0[lo]+1.0E-10);
11     FOR o := 0 TO 15 DO
12         v3[o*28+i] := W_DB.t1[base+lo]*(1-t_val)
13                   + W_DB.t1[base+lo+1]*t_val;
14     END_FOR;
15 END_FOR;
16 END_FUNCTION_BLOCK
```

Line numbers and syntax highlighting rendered as print-ready vector figure.